Yaseen Malam

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Education

Queen's University Kingston, ON
BSc Computing (AI Option), Minor in Economics *Sept 2024 – Apr 2028 (Expected)*

- Coursework: Data Structures & Algorithms, Operating Systems, Machine Learning, Software Engineering

Experience

BMO Financial Group Toronto, ON
Data Analyst Intern, Corporate Audit Division (Audit AI and Analytics) *May 2026 – Aug 2026*

- Built a Python pipeline on AWS SageMaker that matches 250K access records across 3 systems and flags exceptions against 8 audit rules, cutting 65% of the 60,000 hours auditors typically spend on this audit process each year (39,000 hours saved).
- Implemented an AWS Bedrock LLM workflow that turned a manual audit procedure into a reproducible service with fixed prompt templates, reducing preparation and validation effort by 50%; 95% of outputs agreed with auditors' conclusions.
- Automated a multi-stage approval workflow (Power Apps, Power Automate, SharePoint) backing a library of 100+ prompts, saving 5 hours of approval time per item (500 hours saved).

BMC Software Ontario, Canada
Software Engineer Intern *May 2025 – Aug 2025*

- Developed BMC Qubotix: an AI-integrated test automation dashboard that gives QA engineers real-time test case performance and feedback loops, used by 8 QA engineers across 5 test suites.
- Built the full stack: a React/TypeScript frontend and Node.js/Flask REST APIs that surface script modifications and pass/fail rates live, reducing QA feedback turnaround by 35%.
- Orchestrated LLM-driven test script generation with LangChain, cutting manual authoring time by 40%; 85% of 120 generated scripts ran green without edits.
- Integrated Playwright to execute the generated scripts automatically, enabling continuous, hands-off validation of AI-generated tests.

Projects

QU Lecture Library (KingHacks Hackathon Winner) | *TypeScript, React, Fastify, Prisma, SQLite, Python* 2026

- Built a full-stack web app where students upload lecture recordings by course code, then search their course to find any lecture with its AI-generated notes, flashcards and quizzes in one place; tested end to end on 40 lectures.
- Built a Fastify REST API with Prisma/SQLite and Zod validation, documented in OpenAPI, plus a React/Vite/Tailwind frontend.
- Built the upload-to-notes pipeline: ElevenLabs speech-to-text (timestamped transcripts) and an OpenAI note generator via Python/LangChain, with an end-to-end smoke test.

Disaster Tweet Classifier | *PyTorch, Hugging Face Transformers, DistilBERT* 2026

- Fine-tuned DistilBERT (67M parameters) on 7,613 labelled tweets, reaching 81.5% accuracy and 0.81 weighted F1 on 1,142 held-out tweets, versus a TF-IDF + logistic regression baseline of 79%.
- Diagnosed errors with a confusion matrix and per-class analysis: 0.76 recall on the minority class was the dominant failure, and the train/validation loss gap showed overfitting.

Multilingual Patient Intake Assistant | *Node.js, Express, Gemini API, Zod, JavaScript* 2026

- Built a full-stack intake app in English, French, Punjabi and Hindi: patients answer on a phone or tablet, and the doctor sees an AI-organized English summary beside every original answer with its translation.
- Ran summary and translation as separate temperature-0 model calls (Gemini or Claude) with an automated check that flags any medication or number missing from the summary; encrypted data at rest (AES-256-GCM), auto-deleted it after 24 hours, and covered it with 10 automated tests.

Technical Skills

- **Languages:** Python, JavaScript, TypeScript, SQL, Java, C
- **Full-stack:** React, Node.js, Flask, REST APIs, SQLite, Playwright
- **Cloud / Tools:** AWS (SageMaker, Bedrock), Git/GitHub, Jupyter
- **ML / Data:** PyTorch, Hugging Face Transformers, LangChain, scikit-learn, NumPy, pandas